

Table 7: 90% Energy Metrics — All Precision Modes
All values are means over the 28 transformer layers and all component types (q/k/v/o/gate/up/down)

Condition	Prec.	% Params Changed	Rank @90%	σ_k @90%	$\Sigma \sigma_i$ @90%	$\ \Delta W\ _F$	Stable Rank	Spectral Entropy
AdamW+GRPO	FP32	100.0%	190	6.92e-04	0.3321	0.0327	5.4	0.606
AdamW+GRPO	BF16	2.1%	732	1.32e-04	0.1764	0.0063	156.4	0.947
AdamW+GRPO	Mixed	100.0%	329	6.51e-04	0.5112	0.0334	8.3	0.724
Muon+GRPO	FP32	99.9%	752	4.72e-04	0.6200	0.0214	330.1	0.962
Muon+GRPO	BF16	0.5%	590	2.99e-05	0.0339	0.0014	83.0	0.888
Muon+GRPO	Mixed	99.9%	748	4.58e-04	0.6017	0.0209	273.5	0.960
AdamW+SFT	FP32	100.0%	450	7.28e-04	0.7398	0.0387	10.6	0.789
AdamW+SFT	BF16	2.1%	717	3.91e-04	0.5187	0.0187	114.7	0.942
AdamW+SFT	Mixed	100.0%	379	8.55e-04	0.7489	0.0467	7.7	0.720
Muon+SFT	FP32	99.9%	794	7.37e-04	0.9858	0.0333	302.9	0.967
Muon+SFT	BF16	0.4%	492	4.32e-05	0.0402	0.0019	40.6	0.839
Muon+SFT	Mixed	99.9%	730	6.89e-04	0.9023	0.0321	218.4	0.950